Split Capacitor DAC Mismatch Calibration in Successive Approximation ADC

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***Abstract*-A split capacitor DAC calibration method is proposed that a bridge capacitor larger than conventional design allows a tunable capacitor to compensate for mismatch. To guarantee proper calibration, a comparator with digital timing control offset cancellation is proposed. An 8-bit successive approximation ADC with 4b+4b split capacitor DAC calibration has been implemented in 65nm CMOS, achieving 0.3LSB DNL and INL with 180fF input capacitance.**

I. INTRODUCTION

Charge redistribution based successive approximation (SA) ADC has the advantage of low power operation [1-3]. However, input load capacitance and area of a binary-weighted capacitor DAC (CDAC) increase exponentially with the number of bits. Fig. 1(a) shows an 8-bit example. Split CDAC is one of solutions to reduce both input capacitance and area, shown in Fig. 1(b) and Fig. 1(c). In Fig. 1(b), a fractional value bridge capacitor is implemented so that the two capacitor arrays have the same scaling [4]. In the charge redistribution, the total weight of the left array is equal to the weight of the lowest bit in the right array. However, the bridge capacitor being fraction causes poor matching with the other capacitors. In Fig. 1(c), a unit bridge capacitor is implemented and the dummy capacitor C_d is removed [5]. The total weight of the left array achieves the same as the lowest bit in the right array. However, 1LSB gain error is caused. Both the above implementations are vulnerable to parasitic capacitance in node V_Q and between V_Q and V_P, which cause mismatch between the left and right arrays. Since parasitic capacitance is sensitive to process options, such as the number of metal layers, which varies according to system-on-chips (SoC), the split CDAC linearity performance is not well controlled.

This paper proposes a new calibration scheme to improve the linearity performance and avoid gain error as well. As shown in Fig. 1(d), taking parasitic capacitance C_{P1} and C_{P2} into consideration, the bridge capacitor C_B is set to be slightly larger than the ideal fractional value and a tunable capacitor C_C is introduced in parallel with the left array, the lower weight side (L-side), to compensate for mismatches. An extra unit capacitor is added to the right array, the higher weight side (H-side), which will be explained later.

II. PRINCIPLE

The calibration is to adjust the total weight of the L-side capacitor array until it is equal to that of the lowest bit capacitor in the H-side array even if there exist capacitance variation of C_B and parasitic capacitance C_{P1} and C_{P2}. The effective weight of the L-side array can be calculated using the equivalent circuits shown in Fig. 2. Since

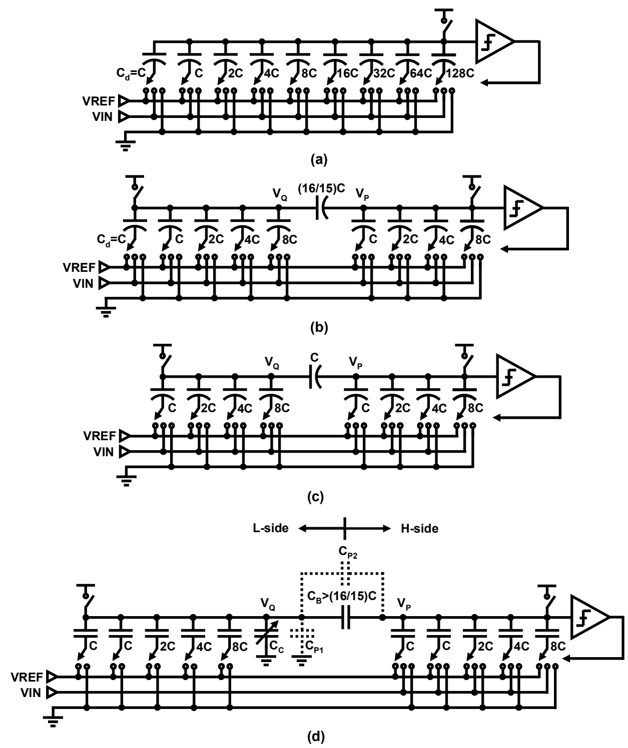

Fig. 1. 8-bit charge redistribution based SA ADC (a) binary weighted capacitor array (b) split array with fractional bridge capacitor (c) split array with unit bridge capacitor (d) proposed split array with mismatch calibration.

the CDAC output node V_P is connected to common mode voltage VCM during sampling phase and returns to VCM in the end of conversion phase, it can be considered a virtual ground. Therefore the parasitic capacitance in node V_P doesn't affect the charge redistribution. Suppose the bottom plates of all L-side capacitors are connected to V_X, according to voltage division principle, V_Q can be calculated as follows.

$$V_Q = \frac{16C}{16C + C_C + C_{P1} + C_B + C_{P2}} \cdot V_X$$

Then the charge contributed to node V_P is as follows.

$$Q = (C_B + C_{P1}) \cdot V_Q = \frac{(C_B + C_{P1}) \cdot 16C}{16C + C_C + C_{P1} + C_B + C_{P2}} \cdot V_X$$

Therefore, the contribution of the total L-side array to node V_P is equivalent to a capacitor with capacitance C_{EF}, which is directly connected to V_P.

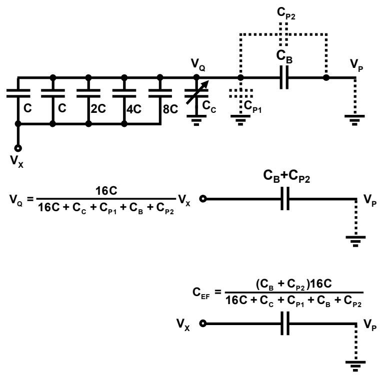

Fig. 2. Equivalent circuits of the L-side capacitor array.

$$C_{EF} = \frac{(C_B + C_{P1}) \cdot 16C}{16C + C_C + C_{P1} + C_B + C_{P2}}$$

Adjust C_C until C_{EF} is equal to the unit capacitor C. It can be easily calculated that the L-side capacitor array has the correct weight in the charge redistribution when C_C=15(C_B+C_{P2})-16C-C_{P1}.

For a more general case, the L-side is a k-bit capacitor array. Its equivalent capacitance to node V_P can be calculated as follows.

$$C_{EF} = \frac{(C_B + C_{P1}) \cdot 2^k C}{2^k C + C_C + C_{P1} + C_B + C_{P2}}$$

When C_C=(2^k-1)·(C_B+C_{P2})-2^k·C-C_{P1}, the L-side achieves the correct weight. In the proposed calibration scheme, both the fractional value bridge capacitor mismatch and the parasitic capacitance effects to linearity performance can be well controlled.

III. Implementation

A. CDAC

The split CDAC calibration implementation with a timing diagram has been shown in Fig. 3. Single-ended circuit is shown for simplicity while fully differential circuit is implemented.

During calibration, all H-side capacitors except for one lowest bit are grounded. When ϕS="1", both nodes V_P and V_Q are biased to VCM. The lowest bit of the H-side array is set to "0" and the L-side array is set to "11111". When ϕS="0", V_P and V_Q become floating nodes. The lowest bit of the H-side array is set to "1" and the L-side array is set to "00000". When C_C is too small, the total weight of the L-side array is larger than the lowest bit of the H-side array. The charge in node V_P has to keep constant and therefore V_P becomes smaller than VCM. The comparator compares V_P and VCM and feeds back its output to control logic to increase C_C so that the weight of the L-side array becomes smaller. The process repeats until V_P is equal to VCM and the comparator output changes. The L-side array has the same weight as the lowest bit of the H-side array and the calibration is finished.

To cover a wide enough calibration range, the bridge capacitor C_B is designed to be 1.4C for the 4b+4b split CDAC implementation. The adjustment range of C_C is 8C and the minimum step is 0.5C. Dummy capacitors around the core CDAC capacitor array have been implemented for symmetry. Making use of these dummy capacitors,

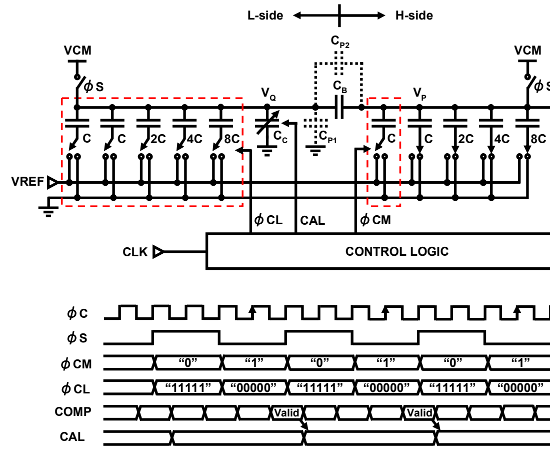

Fig. 3. Split CDAC calibration implementation with timing diagram.

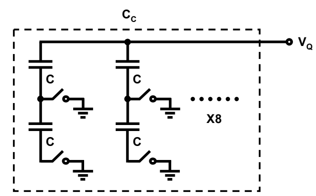

Fig. 4. Implementation of C_C.

no extra area is required to implement C_C. As shown in Fig. 4, two unit capacitors are connected in series to achieve 0.5C step control. When the upper switch turns on, the capacitance is 1C. When the upper switch turns off and the lower switch turns on, the capacitance becomes 0.5C. When both switches are turned off, the capacitance becomes zero. Totally 8 groups of the series capacitors are connected in parallel, working as C_C. Thermometer control is implemented for simplicity.

Correct CDAC calibration requires the comparator offset to be small enough. Therefore, before the CDAC calibration starts, V_P is biased to VCM and comparator offset calibration is performed. This will be discussed in detail later in this section.

Since the capacitor mismatches are due to process variation, the calibration is performed in foreground. After the calibration, the normal ADC function starts. During sampling phase, both nodes V_P and V_Q are biased to VCM. In conventional split CDAC designs, Fig. 1(b) and (c), both H-side and L-side arrays sample the input signal, and therefore the input capacitor load is 31C and 30C, respectively. In the proposed scheme, Fig. 1(d), an extra unit capacitor is added to the H-side. Only the H-side array is used to sample the input. The L-side is grounded during sampling phase. During conversion phase, the extra unit capacitor is grounded while the remaining capacitors are connected to either ground or $VREF$ depending on their respective data bits. Thus, the input load capacitance can be reduced to 16C. In this work, the unit capacitor is set to be 11fF and therefore the total input load is about 180fF. If only the kT/C noise meets the resolution requirement, this method can reduce the input load capacitance into half of the conventional split CDAC.

B. Comparator

A comparator with small enough offset is required to guarantee correct CDAC calibration. Using current sources to compensate for offset causes static power dissipation [6]. Using capacitors to adjust output loads to calibrate offset leads to degradation of speed [7]. This comparator employs a timing control offset calibration technique based on [8]. It has been improved by using a two-step digital control method.

The comparator schematic is shown in Fig. 5. A differential input pair feeds currents into a cross-coupled CMOS regenerative pair. A reset switch *M*1 helps recovering from overdrive as well as allowing offset calibration. Two sets of PMOS transistors, *MC*1 and *MC*2, with source-drain connected are implemented in node *N*1 and *N*2, respectively. Both *MC*1 and *MC*2 consist of coarse and fine calibration transistors *MCC* and *MCF*. *MCC* is controlled by a tunable timing clock *OCC*, while *MCF* is controlled by a fixed timing clock *OCF*.

During the reset phase, two output nodes are pulled up to supply voltage. Node *N*1 and *N*2 are connected through *M*1. *MC*1 and *MC*2 are turned on to save charges in the channels. When the regeneration phase starts, *M*1 is turned off by signal *CLK* and *MCC* on one side is turned off by *OCC* after a delay *td*. Charges in the channel of *MCC* are injected to node *N*1 or *N*2 to generate a compensation voltage. The compensation voltage is determined by the transistor size of *MCC* and the parasitic capacitance in node *N*1 or *N*2. The input-referred offset is amplified by the regenerative pair during the delay period *td*. The longer the delay is, the larger the voltage gain becomes, and therefore the smaller input-referred offset can be compensated. The delay time *td* is controlled by adjusting transistor sizes in an inverter delay line. Fig. 6 shows a simulated relationship between offset and the delay *td*. The coarse calibration covers an offset range of 40mV in 10mV steps.

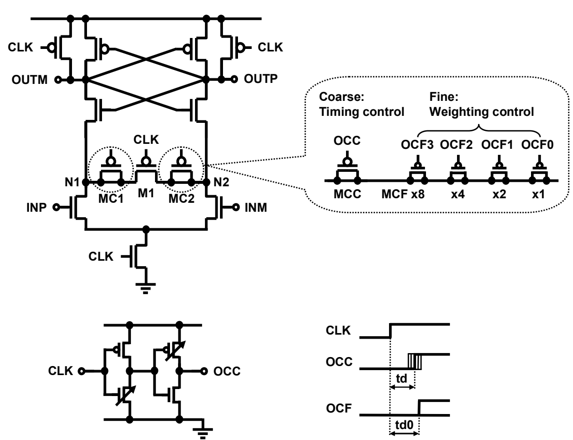

Fig. 5. Comparator with digital timing control offset calibration.

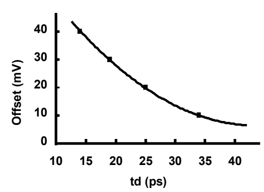

Fig. 6. Simulated relationship between offset and delay time *td*.

Fine calibration is performed by a 4-bit array of smaller size transistor *MCF* applied with a fixed timing clock *OCF*. *MCF* works in a similar way to *MCC*, injecting charges to node *N*1 or *N*2 a delay after *M*1 is turned off. The difference is that the delay period *td*0 is fixed. The compensation voltage generated by the *MCF* array is determined by the number of *MCF* that is chosen to operate. The fine calibration manages to control offset in 1mV steps. The comparator offset calibration needs 20 clock cycles at most, 4 for the coarse calibration and 16 for the fine calibration.

The timing control calibration technique to some extent degrades the comparator speed. However, compared with the calibration using capacitors [7], the timing control technique can achieve the same offset control range with much smaller loading.

IV. Experimental Results

An 8-bit SA ADC with 4b+4b split CDAC calibration has been implemented in a 65nm CMOS process. Fig. 7 shows the chip micrograph. The calibration circuits are implemented on chip. The core area is 0.2mm×0.15mm.

Fig. 8 shows measured static performance of the ADC. The DNL and INL before calibration are shown in Fig. 8(a). The negative periodical peaks in DNL shows that the L-side capacitor array has larger weight than the lowest bit capacitor in the H-side array. INL is the integration of DNL, and therefore peaks with the same period are shown. Fig. 8(b) shows the DNL and INL after the calibration is carried out. They are improved to be +0.2/-0.3LSB and +0.3/-0.3LSB, respectively.

Fig. 9 shows measured SFDR and SNDR as a function of input signal frequency at 50MS/s. The measured ENOB is 7.3 for 20MHz input and 7.1 for 50MHz input. Fig. 10 shows FFT spectrum of the output at 20MHz input.

The ADC power consumption is 1.83mW at 1.2V supply voltage, 0.46mW for analog and 1.37mW for digital. Table I summarizes the ADC performance.

The logic control circuits implemented for the CDAC mismatch calibration and the comparator offset calibration have not been optimized for power consumption and area. The ADC is believed to perform better after circuit optimization. The split CDAC mismatch calibration technique can improve linearity performance more significantly when it is applied to higher resolution SA ADCs.

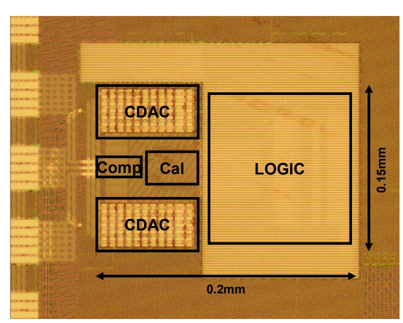

Fig. 7. Chip micrograph.

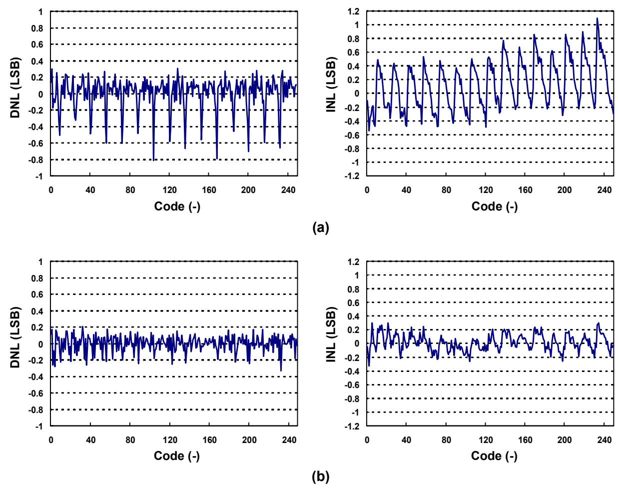

Fig. 8. Measured DNL and INL (a) before calibration (b) after calibration.

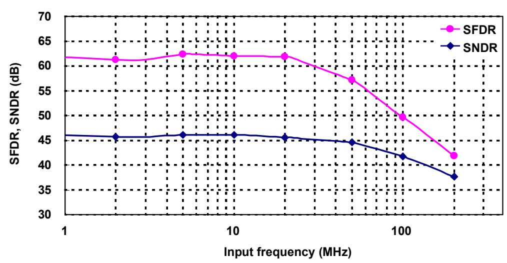

Fig. 9. Measured SFDR and SNDR versus input frequency at 50MS/s.

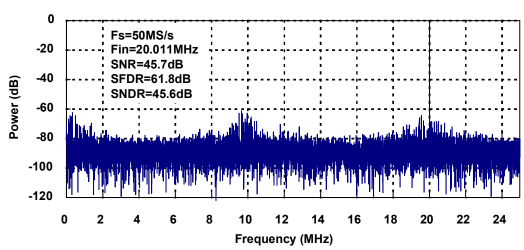

Fig. 10. Measured output FFT spectrum.

V. Conclusion

A new split CDAC mismatch calibration scheme has been proposed. The bridge capacitor is set to be larger than conventional fractional value so that a tunable compensation capacitor can be added to the lower weight side. Comparator offset is cancelled by using a two-step digital timing control method before the CDAC calibration. An 8-bit SA ADC with 4b+4b split CDAC calibration has been implemented in 65nm CMOS. The ADC has achieved +0.2/-0.3LSB DNL and +0.3/-0.3LSB INL with 180fF input capacitance.

TABLE I
Performance Summary

Power supply voltage	1.2V
Input range	1.2 Vp-p differential
Input capacitance	180fF
Sampling rate	50MS/s
Resolution	8 bits
Analog power consumption	0.46mW
Digital power consumption	1.37mW
Total power consumption	1.83mW
DNL	+0.2 / -0.3 LSB
INL	+0.3 / -0.3 LSB
ENOB	7.3 at fin=20MHz 7.1 at fin=50MHz
Area	0.2mm × 0.15mm
Technology	65nm 1P7M CMOS with MIM capacitor

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